

## Table of contents

|                                                                                                                                                            |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Potential as “intrinsic energy of inertia” . . . . .                                                                                                       | 1 |
| Electro-Magnetic Field (force caused by potential) . . . . .                                                                                               | 1 |
| Electro-Magnetic interactions/coupling (Maxwell conditions) . . . . .                                                                                      | 3 |
| Potential . . . . .                                                                                                                                        | 4 |
| elec. Potential structure: . . . . .                                                                                                                       | 4 |
| magn. Potential structure: . . . . .                                                                                                                       | 4 |
| Electric endogen energy . . . . .                                                                                                                          | 4 |
| Electric induced energy (exogenous excitation) . . . . .                                                                                                   | 5 |
| Conductor (charge conductive element for energy transfer) condition: . . . . .                                                                             | 5 |
| Slow changing current (quasi-stationary, low frequencies) condition: . . . . .                                                                             | 5 |
| Energy conservation for (large extended) conductive medium: . . . . .                                                                                      | 6 |
| Energy of thin conductor (concentrated conductive medium without electric charge<br>in its surrounding), Voltage (potentialdifference, tension): . . . . . | 6 |
| Low Frequency . . . . .                                                                                                                                    | 7 |
| High Frequency . . . . .                                                                                                                                   | 7 |
| Continuity at Rest: Electric Charge . . . . .                                                                                                              | 7 |
| Continuity at Movement: Magnetic Field . . . . .                                                                                                           | 7 |
| Continuity at Acceleration: Radiation Energy . . . . .                                                                                                     | 7 |
| Continuity and periodic (harmonious oscillation) Movement in VACUUM . . . . .                                                                              | 8 |
| Continuity and periodic (harmonious oscillation) Movement in MATTER . . . . .                                                                              | 8 |
| Near Field . . . . .                                                                                                                                       | 8 |
| Far Field . . . . .                                                                                                                                        | 8 |
| Dissipation in Vacuum . . . . .                                                                                                                            | 8 |
| Dissipation in Matter . . . . .                                                                                                                            | 9 |
| Energy gain by excitation: . . . . .                                                                                                                       | 9 |

## Potential as “intrinsic energy of inertia”

### Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spatial dependency, no time dependency
  - stationary: source charge at uniform movement, steady current flow, space and time dependency
  - dynamic: source charge at fast movement, radiation, space and time dependency
  - relativistic: source charge at very speedy movement, radiation, space and time dependency
- 
- rest localization vs. time-dependency

- slow vs. fast movement
- vacuum vs. matter
- local vs. global of  $\mathcal{B}$ , singularity vs. regularity of  $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

## Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left( \frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left( c p_{0_{dipol}} + m_0 \right) \quad (1)$$

## Magnetic Field in Matter

Excitation  $\hat{=}$  Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

## Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservative}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[ \underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement (3)}} + \underbrace{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right) \cdot \frac{a}{c}\right]}_{\text{acceleration}} \right]$$

## Electric Field in Matter

Displacement  $\hat{=}$  Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \quad (4)$$

## Abbreviations

$\Phi$  = electric potential       $A$  = magnetic potential       $j = \nabla \times \mathcal{H}$  = elec. current density

$\mathcal{E}$  = Electric Field       $\mathcal{B}$  = Magnetic Field       $\nabla \cdot \mathcal{B} = 0$        $\mathcal{D}$  = Displacement density of dielectricum       $\mathcal{M} =$

$\mathcal{P}$  = Polarisation       $\mathcal{H}$  = magn. excitation, magn. field strength ("magn. displacement")

$n$  = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$        $\chi$  = elec. susceptibility of dielectricum       $\kappa = \frac{\mu - \mu_0}{\mu_0}$  = magn. susceptibility

$\langle p_m \rangle$  = spacial average of magn. dipol moment       $\langle \mathcal{E}_m \rangle$  = spacial average of electric field

$\bar{\mathcal{E}}_m$  = time average of electric field       $\mu$  = magn. permeability       $\mu_0$  = magn. constant ...

$\varepsilon$  = dielectricity constant       $\varepsilon_0 = \dots$

## Electro-Magnetic interactions/coupling (Maxwell conditions)

**conservativ, static charge localisation:**

...

**stationary, slow charge movement:**

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \qquad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

**relativistic, fast charge movement:**

...

## Potential

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomogeneity) from source at rest,  $v = 0$ :

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

**elec. Potential structure:**

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

**magn. Potential structure:**

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

## Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

### Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2}mv^2 \hat{=} (pc)^2 \hat{=} h\nu$$

$E_{kin}$  Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2 \quad (12)$$

$E_{kin}$  Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \quad (13)$$

$E_{kin}$  Electric Energy:

$$W_e = \frac{1}{2}CU^2 \quad (14)$$

$E_{kin}$  Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

### Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (17)$$

### Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (18)$$

**Energy conservation for (large extended) conductive medium:**

$$\int \left( \frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left( \frac{Q^2}{C} \right) \quad (19)$$

**Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):**

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (20)$$

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogeneity, cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \quad C = \text{Capacitance} \quad Q = \text{charge (ion)} \quad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{radiation intensity}$$

$$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr = \text{dissipation energy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \hat{=} kc \hat{=} \frac{1}{\sqrt{LC}} \hat{=} \sqrt{\frac{g}{r}} = \text{frequency}$$

$\mu$  = permeability of conductor       $\sigma$  = conductivity of conductor (dielectricum)

### Low Frequency

$$\omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \quad (21)$$

$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

### High Frequency

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (22)$$

$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

### Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

### Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

### Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

## Continuity and periodic (harmonic oscillation) Movement in VACUUM

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

## Continuity and periodic (harmonic oscillation) Movement in MATTER

Refraction index emerges

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

### Near Field

$$r \ll \lambda = \frac{2\pi}{k} \text{ vacuum} = \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

### Far Field

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \text{ vacuum} = \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left( \mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\varepsilon_0}{2} c^2 \left( \mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

### Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

## Dissipation in Matter

no Continuity of charge/matter with Dielectricum, Refraction merges, Reflection, Absorption

## Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)<sup>1</sup>:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (23)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (24)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (25)$$

$V_0$  = Opposing Potential ( $I = 0$ )

$\alpha = \angle(E_{kin}, \nu)$       $e$  = electron charge      $Q$  = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left( \frac{e}{h} Q + \nu_0 \right)^2$$

$\Rightarrow$  Ionisation Energy of chemical element  $E_i = h\nu = h \frac{\lambda}{c} = \hbar \omega = \frac{h}{k\lambda} \omega$

$\Rightarrow$  Energy released (gained)  $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

---

<sup>1</sup>Wirkungsquantum